

Assignment No. - 4

ON

SSCA4030 - Image Processing

TITLE: Apply spatial filters: mean, median, Gaussian, Laplacian.

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

SUBMITTED BY:

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Input:

```
import cv2
from matplotlib import pyplot as plt

# Read grayscale image
img = cv2.imread("/content/img1.jpg", cv2.IMREAD_GRAYSCALE)

# Apply a 3×3 Mean (Average) Filter on the input image
# Mean Filter
mean = cv2.blur(img, (3, 3))

# Median Filter
median = cv2.medianBlur(img, 3)

# Apply a 3×3 Gaussian Filter
# Gaussian Filter
gaussian = cv2.GaussianBlur(img, (3, 3), 0)

# Laplacian Filter (with larger kernel size)
laplacian = cv2.Laplacian(img, cv2.CV_64F, ksize=5)
laplacian_abs = cv2.convertScaleAbs(laplacian)

# Apply threshold to make edges more visible
# The threshold value (50) may be adjusted depending on the image
_, laplacian_thresholded = cv2.threshold(
    laplacian_abs,
    50,
    255,
    cv2.THRESH_BINARY
)

# Display Results
titles = [
    "Original",
    "Mean Filter",
    "Median Filter",
    "Gaussian Filter",
    "Laplacian Filter (Abs, ksize=5)",
    "Laplacian Filter (Thresholded, ksize=5)"
]

images = [
    img,
    mean,
    median,
    gaussian,
    laplacian_abs,
    laplacian_thresholded
]
```

```
plt.figure(figsize=(15, 10))

for i in range(len(images)):
    plt.subplot(2, 3, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Original



Mean Filter



Median Filter



Gaussian Filter



Laplacian Filter (Abs, ksize=5)



Laplacian Filter (Thresholded, ksize=5)

